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Cis 130: Computer Organization

Assignment 4: Combinational Circuits

1) Truth tables for $O_1 - O_7$

Create seven truth tables, one for each output signal.

O_1)

	I_1	I_0	Disable							
	A	B	C	O_1	O_2	O_3	O_4	O_5	O_6	O_7
"0"	0	0	0	1	1	1	1	1	1	0
"1"	0	1	0	0	1	1	0	0	0	0
"2"	1	0	0	1	1	0	1	1	0	1
"3"	1	1	0	1	1	1	1	0	0	1
	0	0	1	0	0	0	0	0	0	0
	0	1	1	0	0	0	0	0	0	0
	1	0	1	0	0	0	0	0	0	0
	1	1	1	0	0	0	0	0	0	0

off.

O_1	
O_6	O_2
O_3	O_7
O_4	O_5

0	0	0	0
0	1	1	1
1	0	2	2
1	1	3	3

2) Boolean SOP Expressions

for each output: O_i :

A) find all rows where $O_i = 1$

B) Write the product terms for those rows.

C) Combine them to form the SOP expression

$$\begin{aligned}
 O_1 &= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + AB\bar{C} \\
 O_2 &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} \\
 O_3 &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} \\
 O_4 &= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + AB\bar{C} \\
 O_5 &= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} \\
 O_6 &= \bar{A}\bar{B}\bar{C} \\
 O_7 &= A\bar{B}\bar{C} + AB\bar{C}
 \end{aligned}$$

3) Logic Circuit Diagram.

for each output: O_i :

A) Draw the circuit that implements each output using AND, OR and NOT

